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Runtime Validity Estimation for the Integration of Hybrid Learned Models into Vehicle Dynamics Control

Doctoral Candidate

Supervisor

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Abstract

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Chapter 1

Introduction

Consider a driver traveling along a highway at 100 km/h when it starts to rain. As visibility decreases, the driver decides to pull over onto the shoulder. During the maneuver, however, water on the road may reduce tire–road grip, making it harder to change course. What began as an attempt to stop the journey can, under these conditions, lead to a loss of control and a risk of collision with other vehicles or roadside barriers.

This example highlights a difficulty that extends beyond the driver's decision: the ability to execute a maneuver depends on the contact conditions between the tires and the road. When these conditions change, the forces available for braking or changing direction may also change. Understanding how grip varies along the route is therefore part of the problem of keeping the vehicle on its intended path.

Pavement wear and water on the road are among the factors that affect grip. Over time, traffic polishes the aggregates and changes the surface texture, reducing skid resistance [17, 8, 18]. During rainfall, water can impair tire–road contact and, under hydroplaning conditions, cause that contact to be lost [13]. Ruts in the wheel paths allow water to accumulate, contributing to the problem [12]. Both pavement wear and the conditions encountered during a journey can therefore alter the grip available for a maneuver.

This limitation becomes relevant when a driver needs to change lanes at short notice or swerve to avoid an obstacle. In these maneuvers, the lateral forces required to follow the intended path may exceed those that the tires can transmit to the road [1]. The combination of high speed and low grip can then cause lateral sliding and yaw instability, leading to a loss of vehicle control [7]. The challenge is therefore to execute the maneuver within the limits imposed by tire–road interaction. Road conditions are one source of variation; changes in vehicle loading and tire properties can also alter the response to braking and steering inputs.

Vehicle control systems address this challenge by assisting the driver with braking and maintaining vehicle stability. The anti-lock braking system (ABS) and electronic stability control (ESC) illustrate this development: the former aims to prevent wheel lock during braking, while the latter helps correct deviations between vehicle motion and the driver's

intended direction. The introduction of ESP, an implementation of electronic stability control launched by Bosch in 1995, extended this assistance to situations involving a loss of stability [3]. Development continues through the integration of braking, steering, and propulsion [14].

Coordinating these systems expands the controller's ability to influence vehicle motion. Their effectiveness, however, depends on accounting for variations in both operating conditions and vehicle parameters, including quantities that are not directly measured during operation.

1.1 Context and motivation

The opening example illustrates how road conditions can alter the vehicle's response, but similar concerns arise when the vehicle itself changes. Carrying passengers or cargo changes its mass and loading, while tire properties influence its response to steering. These variations matter for different aspects of motion: friction and axle loads affect braking-distance prediction [22], whereas mass and tire cornering stiffness enter the description of lateral and yaw dynamics [21]. Cornering stiffness describes the relationship between lateral tire force and slip angle within the linear range; it serves a different role from the friction limit that bounds the available tire forces. Accounting for these quantities therefore involves both the forces the vehicle can generate and the motion that results from them.

Studies of adaptive vehicle control provide examples of how these variations affect tracking performance. Xu et al. [19] evaluate trajectory tracking in simulations on roads with low and changing friction. In the scenarios considered, their adaptive strategy reduces lateral and heading errors relative to the comparison controllers when grip changes along the path. Yeneneh et al. [21] investigate lateral tracking under variations in vehicle mass, speed, and friction, using online parameter estimation to update the controller's prediction model. Their simulations report lower tracking errors than conventional MPC and LQR controllers under the conditions evaluated. Their formulation also includes the estimation of cornering stiffness, illustrating how parameters that are not directly measured can be inferred from the vehicle's response.

For this thesis, these examples motivate the study of changes in operating conditions and vehicle parameters, including friction, mass and loading, and tire cornering stiffness. A parameter may be uncertain even when it remains constant, and a physical change may occur between journeys or during one. The controller must work with the measurements, estimates, and nominal values available to it. This raises the question of how well its model continues to describe the vehicle as those conditions and parameters vary, providing the starting point for examining model mismatch and its consequences for control.

Controllers that predict vehicle motion use a model to relate their commands to the

response they expect. Model predictive control (MPC), including its nonlinear formulation (NMPC), uses these predictions to choose control inputs over a finite horizon while accounting for constraints [15]. The model must describe the dynamics in enough detail to support those decisions while allowing the control problem to be solved within the available computation time. Chen et al. [5], for example, use a simplified bicycle model to balance prediction accuracy and computational cost. Pan et al. [11] identify model structure and parameter errors as sources of discrepancy between a control-oriented model and the vehicle. These discrepancies, referred to as model mismatch, can affect predictions and control performance even when the optimization problem is solved as intended.

Parameter uncertainty adds to the mismatch introduced by model simplification. Yeneneh and Sufe [20] consider uncertainty in vehicle mass, front and rear tire cornering stiffness, and road friction within a robust predictive-control formulation. Li et al. [10] address uncertainty in cornering stiffness and yaw moment of inertia in a controller that includes longitudinal and lateral coupling, evaluated through hardware-in-the-loop tests. These studies illustrate why the issue extends beyond changes in grip: the vehicle's response also depends on parameters that may differ from the values used in its model. Structural approximations and parameter uncertainty can therefore contribute to prediction error through different mechanisms.

It is useful here to distinguish the physical parameters of the vehicle, the nominal values chosen for its model, and any estimates obtained from measurements during operation. An estimate provides information about the vehicle, but may still differ from the physical value; calibrating parameters also leaves discrepancies associated with the model's structure. The formulation of Chen et al. [4] reflects this distinction by combining calibration of a nominal physical model with compensation for remaining unmodeled dynamics. For this thesis, the question is how those discrepancies affect the information used by the controller and how to improve its predictions without losing the computational advantages of a simplified model.

Hybrid modeling offers a way to address part of this mismatch by combining physical equations with information learned from data [6, 23]. In a model with an additive learned residual, the physical component provides a nominal prediction of vehicle motion, and the learned component estimates a correction from the difference between that prediction and the observed response. The residual can thus capture effects that the simplified model does not describe adequately, while the physical equations remain part of the prediction model [9, 5]. In Kabzan et al.'s formulation, the training targets are obtained from measured state transitions and nominal predictions, allowing the model error to be learned without identifying its physical causes separately [9].

Kabzan et al. demonstrate this approach in autonomous racing, using Gaussian process regression to learn an additive correction to a nominal vehicle model. In tests on a full-size race car, the corrected model maintained an approximately constant average

prediction error as driving became more aggressive over successive laps, while the error of the nominal model increased [9]. Hybrid models can also assign different roles to learning and physical estimation. Zhou et al., for example, combine a neural network model of traction with a physical longitudinal dynamics model and a parameter estimator for vehicle mass and a modified rolling friction coefficient. Their evaluation includes changes in load and road conditions beyond those used for training [23]. This architecture illustrates how learned force models and parameter estimation can complement physical equations, although it differs from an additive residual model.

Other hybrid approaches fuse physical and neural models for dynamic prediction or couple a neural estimator with a physical observer for sideslip estimation [6, 2]. Physical knowledge can also guide model training: Wang et al. incorporate residuals of force-equilibrium equations into the learning objective of a vehicle dynamics model [16]. These examples show that physical knowledge and learning can be combined in different ways. The formulation adopted in this thesis focuses on an additive learned residual within the controller's prediction model, whose benefit will be assessed through prediction accuracy and its effect on control. Retaining physical equations provides structure, but it does not establish that the learned correction remains useful under every operating condition. Chen et al. identify the limited adaptability of an offline-trained correction to new conditions as a motivation for combining offline learning with online updates [5]. This dependence on the conditions represented by the data leads to the question of how to assess the correction's validity during operation.

1.2 Research problem

1.3 Scope of the problem (Priority 1)

1.4 Research gap

1.5 Research question

How can the runtime validity of a learned vehicle dynamics residual model be estimated and used to adapt an MPC/NMPC controller under operating conditions not represented during training?

1.6 Specific research questions

RQ1: Which types of physically relevant shifts in vehicle dynamics cause a significant loss of validity in the learned residual?

RQ2: What information do epistemic uncertainty, aleatoric uncertainty, calibration, and an OOD indicator provide about control-relevant model error?

RQ3: To what extent are uncertainty, calibration, and OOD signals complementary or redundant in characterizing runtime validity?

RQ4: How can a confidence/validity signal $C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Calibration}, \text{OOD})$ be constructed to reflect the error relevant to the controller?

RQ5: How should C_k be used to adapt the MPC/NMPC horizon to reduce closed-loop performance degradation when the validity of the learned model decreases?

RQ6: What improvement does the proposed architecture offer over a controller based only on the physical model, a controller that uses the learned model without validity monitoring, and architectures that use only validity or only OOD information?

RQ7: As an optional extension, can validity information be used to modify the action of a CBF or trigger data/model update strategies in response to epistemic uncertainty?

1.7 Research hypothesis

A calibrated combination of epistemic uncertainty, aleatoric uncertainty, and evidence of out-of-distribution operation can provide a useful estimate of the runtime validity of a learned vehicle dynamics residual. Incorporating this estimate into the adaptation of an MPC/NMPC controller can reduce closed-loop performance degradation under regime changes compared with architectures that use the learned model without validity awareness.

Chapter 2

Objectives

2.1 Overall objective

Develop and evaluate a methodology for real-time validity estimation for a hybrid vehicle dynamics model comprising a physical model and a learned residual, and integrate this estimate into an MPC/NMPC controller to reduce closed-loop performance degradation under operating conditions not represented during training.

2.2 Specific objectives

SO1: Formulate a hybrid vehicle dynamics architecture comprising a control-oriented physical model and a learned residual.

SO2: Provide operational definitions of the types of shift/OOD relevant to the study, distinguishing at least unrepresented physical regimes and dynamics OOD, and including input OOD if time permits.

SO3: Estimate and calibrate the aleatoric and epistemic uncertainty of the learned model using a heteroscedastic model and a small ensemble, complemented by a computationally inexpensive OOD indicator.

SO4: Experimentally characterize the relationships among distribution shift, uncertainty, calibration, OOD, prediction error, and control-relevant error.

SO5: Define a representation of runtime confidence/validity

$$C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Calibration}, \text{OOD})$$

and evaluate its ability to anticipate or represent model degradation in a way that is useful to the controller.

SO6: Integrate the validity signal into an MPC/NMPC controller through horizon adaptation and evaluate its effect on closed-loop performance and empirical robustness.

SO7: Compare the proposed approach with baseline architectures that isolate the effects of the learned model, validity estimation, and OOD detection.

SO8: As an extension, if the schedule permits, evaluate the use of the validity signal in a CBF and/or a DAgger-based data acquisition and retraining strategy for conditions associated with epistemic uncertainty.

Chapter 3

Fundamentals of vehicle dynamics and control

3.1 Vehicle dynamics relevant to control

3.2 Control-oriented models and model mismatch

3.3 Hybrid models: physics and learned residuals

3.4 Model predictive control

3.5 Control barrier functions as an optional extension

Chapter 4

Learned information and runtime validity

4.1 Learned information relevant to the controller

4.2 Predictive uncertainty

4.2.1 Aleatoric uncertainty

4.2.2 Epistemic uncertainty

4.2.3 Calibration

4.3 OOD and distribution shift in vehicle dynamics

4.3.1 Unrepresented physical regime

4.3.2 Dynamics OOD

4.3.3 Input OOD

4.3.4 Sensor failure or degradation

4.4 Runtime validity and confidence

$$C_k = g(U_{\mathrm{epi},k}, U_{\mathrm{ale},k}, \mathrm{Cal}_k, S_{\mathrm{OOD},k}) \,, \tag{4.1}$$

4.5 Control-relevant validity

Chapter 5

State of the art

5.1 Literature review strategy

5.2 Learned and hybrid models for vehicle dynamics control

5.3 Uncertainty-aware learned dynamics for control

5.4 OOD, distribution shift, and runtime monitoring

5.5 Validity-aware and confidence-aware control

5.6 Critical synthesis and research gap

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Conceptual framework and proposed architecture

6.1 The shift–validity–control chain

6.2 Hybrid prediction model

6.3 Validity estimator

6.4 Integration with MPC/NMPC

6.5 Optional extensions

6.5.1 Modulation of a CBF

6.5.2 Data acquisition and retraining

Chapter 7

Research and validation methodology (Priority 1)

7.1 Overall strategy

S1: Establish the control-oriented physical model and the baseline MPC/NMPC controller.

S2: Train the learned residual and quantify its improvement and limitations relative to the physical model within the nominal domain.

S3: Implement aleatoric and epistemic uncertainty estimation, calibration, and a computationally inexpensive OOD score.

S4: Design physically interpretable shifts in vehicle dynamics and characterize their effects on prediction, uncertainty, OOD, and validity.

S5: Construct and validate the signal C_k against control-relevant error.

S6: Integrate C_k into MPC/NMPC adaptation.

S7: Compare the baselines and the proposed approach under nominal and OOD conditions.

S8: Evaluate empirical robustness and, if progress permits, incorporate CBF and/or DAgger extensions.

7.2 Distribution shift scenarios

7.2.1 Unknown change in friction

7.2.2 Extension of the speed range

7.2.3 Sensor failure or degradation

7.2.4 Input OOD

7.3 Learned model and uncertainty

7.4 Construction and evaluation of the validity signal

7.5 Validity-aware integration with MPC/NMPC

7.6 Baselines and ablation study

B0 – Physics-only: Controller using a physical model without a learned residual.

B1 – Learned/no-awareness: Controller using a physical model and a learned residual, without validity estimation or OOD detection.

B2 – Validity/uncertainty-only: Controller using a learned model and a signal constructed from uncertainty/calibration, excluding an explicit OOD component.

B3 – OOD-only: Controller using a learned model and a response based solely on the OOD score, without probabilistic uncertainty estimation.

P – Proposed: Controller using a physical model, a learned residual, and the signal $C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Cal}, OOD)$, used to adapt the MPC/NMPC controller.

7.7 Evaluation metrics

7.7.1 Model metrics

7.7.2 Uncertainty, calibration, and OOD metrics

7.7.3 Closed-loop metrics

7.8 Empirical robustness criterion

$$\Delta J_{\text{prop}} < \Delta J_{\text{baseline}}, \tag{7.1}$$

7.9 Level of assurance and scope of the conclusions

7.10 Statistical analysis and reproducibility

Chapter 8

Preliminary results

8.1 Feasibility of learning vehicle dynamics models

8.2 Uncertainty and OOD behavior under distribution shifts

8.3 Preliminary integration with control

8.4 Discussion of preliminary results

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Expected contributions

9.1 Control-relevant validity estimation

9.2 Validity-aware vehicle control

9.3 Runtime validity characterization

9.4 Experimental taxonomy of shifts in vehicle dynamics

9.5 Potential extensions

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Work plan (Priority 1)

10.1 Remaining activities

10.2 Schedule

10.3 Decision milestones

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11.1 Risk of redundancy between uncertainty and OOD

11.2 Risk of poor calibration or weak correlation with error

11.3 Risk that horizon adaptation does not improve closed-loop performance

11.4 Computational risk

11.5 Risk of scope expansion

11.6 Limitations of formal guarantees

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Conclusions of the qualifying examination

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